

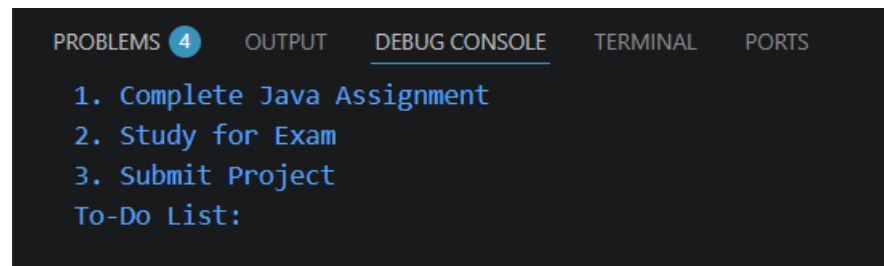


CSEA2

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 2 J AS4EX2.java 3
```

```
J AS5EX1.java > 🐞 TodoList > 📁 main(String[] args)
1 import java.util.ArrayList;
2
3 public class TodoList {
   Run main | Debug main
   public static void main(String[] args) {
4
5       ArrayList<String> tasks = new ArrayList<>();
6
7       // Adding tasks
8       tasks.add("Complete Java Assignment");
9       tasks.add("Study for Exam");
10      tasks.add("Submit Project");
11
12      StringBuffer sb = new StringBuffer();
13      sb.append("To-Do List:\n");
14
15      for (int i = 0; i < tasks.size(); i++) {
16          System.out.println((i + 1) + ". " + tasks.get(i));
17      }
18
19      System.out.println(sb);
20
21  }
22 }
```

OUTCOME-



2. Create a Student Course Registration System using ArrayList to store the list of courses registered by a student and StringBuffer to generate and display the registered course list. The application should allow users to add, remove, and view registered courses.

```
J AS3EX1.java J AS3EX2.java 1 J AS4EX1.java 2 J AS4EX2.java 3 X va 3 X Untitled-1
J AS3EX2.java > CourseRegistration
1 import java.util.ArrayList;
2 import java.util.Scanner;
3
4 public class CourseRegistration {
5     Run main | Debug main
6     public static void main(String[] args) {
7
8         Scanner sc = new Scanner(System.in);
9         ArrayList<String> courses = new ArrayList<>();
10
11         int choice;
12
13         do {
14             System.out.println("\n1. Add Course");
15             System.out.println("2. Remove Course");
16             System.out.println("3. View Courses");
17             System.out.println("4. Exit");
18             System.out.print("Enter Choice: ");
19             choice = sc.nextInt();
20             sc.nextLine();
21
22             switch (choice) {
23                 case 1:
24                     System.out.print("Enter Course Name: ");
25                     String course = sc.nextLine();
26                     courses.add(course);
27                     System.out.println("Course Added.");
28                     break;
29
30                 case 2:
31                     System.out.print("Enter Course Name to Remove: ");
32                     String removeCourse = sc.nextLine();
33                     courses.remove(removeCourse);
34                     System.out.println("Course Removed.");
35                     break;
36
37                 case 3:
38                     StringBuffer sb = new StringBuffer();
39                     sb.append("Registered Courses:\n");
40
41                     for (String c : courses) {
42                         sb.append(c).append("\n");
43                     }
44
45                     System.out.println(sb);
46                     break;
47
48                 case 4:
49                     System.out.println("Exiting...");
50                     break;
51
52                 default:
53                     System.out.println("Invalid Choice!");
54             }
55
56         } while (choice != 4);
57
58         sc.close();
59     }
60 }
```

OUTCOME-

```
PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice:
→ 1

Enter Course Name:
→ ENGLISH

Course Added.

1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice:
→ 3

Registered Courses:
ENGLISH
2
1. Add Course
2. Remove Course
3. View Courses
4. Exit
Enter Choice:
→ 4

Exiting...
```

CONCLUSION-

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The **To-Do List application** was successfully developed using **ArrayList** and **StringBuffer**. The **ArrayList** was used to store and manage tasks dynamically, allowing easy addition and retrieval of tasks. The **StringBuffer** was used to efficiently create and display the task list by modifying strings without creating multiple string objects. This practical demonstrated the effective use of Java Collection classes and mutable strings for dynamic data storage, management, and efficient string manipulation.